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Non-Parametric Learning Assignment 1

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Abstract

The study focuses on optimizing 3 types of feed-forward neural networks across six datasets, three for classification and three for regression, by tuning parameters for each dataset. The 3 types of networks are a single layer network, a multilayer network, and an autoencoder. Performance of the models is evaluated against a null model, either the mode of the class or the mean of the value, as well as by the total loss of the network for a testing dataset. Results demonstrate that linear models outperform the null model for 5 of 6 datasets, multilayer models outperform for 4 of 6, and appended autoencoders outperform for 5 of 6. The study's findings underscore the effectiveness of neural network parameter tuning and autoencoding in enhancing model performance across diverse datasets.

1 Problem Statement

We seek to tune neural networks on 6 different datasets, 3 for classification and 3 for regression. For the single layer networks, we seek to tune the training rate η. For the multilayer networks, we seek to tune the number of nodes in the 2 hidden layers, N1 and N2. For the autoencoder, we seek to tune the number of nodes in the hidden layer of the autoencoder, N1, and then the number of nodes in a hidden layer appended to the trained autoencoder, N2.

2 Pre-Processing Steps

We download the 6 datasets from the Introduction to Machine Learning class page and prepare the data for calibrating the models. For this preparation, we assign headers as dictated from the data descriptions, if necessary; we identify the class or value to be predicted; and we classify each factor as numerical or categorical. Ordinal features, specifically the Month and Date features of the Forest Fires dataset, are treated as categorical.

Most of the datasets contain missing values, denoted by NaN or '?'. An exception is the House Votes 84 dataset, in which '?' signifies abstention from a vote. Thus, we address these NaN and '?' values in all of the datasets except House Votes 84. For numerical factors, we set these missing values equal to the average value of that factor for the full dataset; for categorical factors, we set these missing values equal to the mode of that factor for the full dataset.

Next, we assign a unique sample number to each row in the dataset to act as a primary key on each dataset and allow for easy identification of individual samples.

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We then perform dataset-specific processing. For the Forest Fires dataset, we address the clustering near 0 of the Area predictor by apply a log function function. For the Machine dataset, we remove the columns for vendor name, model name, and ERP, which are not to be used as predictive factors. Finally, for Breast Cancer Wisconsin, we remove the Sample code number column.

For all 6 datasets, we transform categorical features using one hot coding, dropping the first category from each so as to avoid redundancy.

Finally, we randomly split the dataset into a training set and a tuning set, with the tuning set comprising 20% of the provided samples. For the 3 classification datasets, we stratify by the predictor class in this split to ensure that 20% of each class is part of the tuning set.

3 Experimental Approach and Assumptions

To train each network, we utilize a variant of Stochastic Gradient Descent whereby we randomize the order of the training set, feedforward each sample, and backpropagate the calculated error on the result to update the network's weights. Once all samples have been applied, we randomize the order and start again. Performance on the network is checked after every 2 runs through the training data by calculating the total loss function of classifying/regressing the samples in a tuning set. For classification, loss is cross entropy loss; for regression, loss is mean squared error; and for autoencoding, loss is reconstruction loss. Training continues until loss does not decrease at least 1% between checks. This stopping criterion ensures training continually improves performance without running unnecessarily long, and this SGD approach ensures all samples in the dataset help train the model. Finally, the overall quality of the

To tune η, values of .0001, .001, .01, and .1 are tested for all datasets' linear models. These different magnitudes are necessary to capture the different magnitudes of weights required for the various regression and classification problems. E.g. the values to predict for the Forest Fires dataset are far smaller than those for the Machine dataset. A dataset's top performing η is subsequently used to train its multilayer and autoencoder networks.

To tune N1 and N2 for the multilayer networks, multiples of the number of features in the dataset are used, inclusive of the one-hot coding features. Specifically, candidate values for N1 and N2 are determined by multiplying the number of features by 1/4, 1/2, 1, 2, and 4. E.g. for the Car dataset, which, when incorporating the one hot coding categories, contains 15 samples, values of 3, 7, 15, 30, and 60 are tested. The top 2 or 3 performing N2 values are used to tune the Appended Autoencoder networks.

Finally, for the autoencoder, a similar approach to the multilayers' is taken, but multipliers of 1/4, 1/2, and 3/4 are applied to the numbers of features. Once appended to the multilayer network, entire network is trained as a complete feedforward network, allowing for the fine tuning of the weights feeding the encoding layer from the autoencoder as well as learning the weights in the prediction layer.

4 Results

N.B. In the tables below, the 'Samples' column refers to the average number of samples used to train the network. Additionally, certain N1 and N2 combinations are omitted for the sake of space.

4.1 Breast Cancer Wisconsin

Networks trained with all 4 learning rates outperform the Null Model in terms of hit rate. Cross entropy loss is minimized by an η of .1. All significantly outperform the null model.

Table 1: Breast Cancer Wisconsin Linear Model

η	Samples	Loss	Hit Rate	Null Model
0.000100	7840	-226.506987	0.966047	0.655359
0.001000	3360	-254.086809	0.967473	0.655359
0.010000	1680	-263.031691	0.962836	0.655359
0.100000	1232	-265.142086	0.961410	0.655359

Besides the models with 2 nodes in both hidden layers, the multilayer models for Breast Cancer Wisconsin all perform about the same, with cross entropy loss of about -260 and hit rate of about 96%. All significantly outperform the null model.

Table 2: Breast Cancer Wisconsin Multilayer Model

η	N1	N2	Samples	Loss	Hit Rate	Null Model
0.100000	2	2	2186	-207.592864	0.808545	0.655359
0.100000	2	4	2968	-262.971283	0.962137	0.655359
0.100000	2	9	2856	-263.224895	0.962132	0.655359
0.100000	2	18	2800	-262.694916	0.962135	0.655359
0.100000	2	36	2575	-258.759773	0.961428	0.655359
0.100000	4	2	2687	-250.758433	0.948520	0.655359
0.100000	4	4	2744	-262.988080	0.962496	0.655359
0.100000	4	9	2576	-263.593585	0.961776	0.655359
0.100000	4	18	2464	-263.691108	0.962491	0.655359
0.100000	4	36	2408	-263.322531	0.963207	0.655359
0.100000	9	2	2688	-252.292446	0.932951	0.655359
0.100000	9	4	2520	-263.248854	0.961769	0.655359
0.100000	9	9	2296	-263.519956	0.963917	0.655359
0.100000	9	18	2351	-264.117610	0.964628	0.655359
0.100000	9	36	2351	-263.743204	0.960360	0.655359
0.100000	18	2	2800	-262.994187	0.963207	0.655359
0.100000	18	4	2519	-263.799631	0.965699	0.655359
0.100000	18	9	2296	-264.027078	0.963917	0.655359
0.100000	18	18	2295	-264.371338	0.964273	0.655359

Similar to the multilayer networks, the autoencoders perform consistently across η and N1. Attaching to a network, performance remains around -263 cross entropy loss and 96% hit rate.

Table 3: Breast Cancer Wisconsin Autoencoder

η	N1	Samples	Loss
0.010000	2	4480	-254.413590
0.010000	4	4480	-257.479697
0.010000	6	4480	-258.082956
0.100000	2	1680	-263.512700
0.100000	4	1680	-264.141456
0.100000	6	1680	-264.518640

Table 4: Breast Cancer Wisconsin Autoencoder Appended to Network

η	N2	Samples	Loss	Hit Rate	Null Model	Autoencoder η	Autoencoder N
0.100000	4	4872	-263.476737	0.961779	0.655359	0.100000	4
0.100000	9	4647	-263.710180	0.960709	0.655359	0.100000	4

4.2 House Votes 84

Performance of the linear model improves as η increases, as fewer samples are required for training, hit rate increases, and loss decreases. All significantly outperform the null model.

Table 5: House Votes 84 Linear Model

η	Samples	Loss	Hit Rate	Null Model
0.000100	5688	-121.847464	0.889984	0.613186
0.001000	2862	-144.681551	0.911202	0.613186
0.010000	1675	-156.954492	0.940420	0.613186
0.100000	1012	-161.906157	0.944989	0.613186

The multilayer models perform best with a first hidden layer of size 16 or 32 and a second layer of size 8, 16, or 32. All significantly outperform the null model.

The autoencoders similarly perform better at higher η's, with performance varying little for different N1's. Attaching to a classification model, N2 of 8 or 16 outperforms N2 of 32, and overall the models only slightly under-perform the basic multilayer models.

4.3 Car

Performance of the linear models for the Car dataset improves as η increases, as fewer samples are required for training, hit rate increases, and loss decreases. All models besides those trained with η of .0001 outperform the null model.

Table 6: House Votes 84 Multilayer Model

η	N1	N2	Samples	Loss	Hit Rate	Null Model
0.100000	8	8	1710	-139.896333	0.841947	0.613186
0.100000	8	16	1821	-146.033248	0.851093	0.612990
0.100000	8	32	1860	-148.421788	0.871403	0.612990
0.100000	8	64	1898	-152.685959	0.882686	0.612990
0.100000	16	8	2248	-162.531887	0.954176	0.612990
0.100000	16	16	1899	-161.294269	0.944583	0.612990
0.100000	16	32	1899	-163.512660	0.956085	0.612990
0.100000	16	64	2054	-162.870774	0.949688	0.612990
0.100000	32	8	2248	-162.080958	0.952246	0.612990
0.100000	32	16	2054	-162.982350	0.959883	0.612990
0.100000	32	32	1899	-162.321127	0.952253	0.612990
0.100000	32	64	1976	-161.001143	0.938241	0.612990
0.100000	64	8	2209	-161.747480	0.947801	0.612990
0.100000	64	16	2015	-160.985791	0.946506	0.612990
0.100000	64	32	2054	-161.773420	0.949053	0.612990
0.100000	64	64	1860	-161.879844	0.944609	0.612990

Table 7: House Votes 84 Autoencoder

η	N1	Samples	Loss
0.010000	8	2335	-116.063949
0.010000	16	1501	-104.249666
0.010000	24	1779	-111.110699
0.100000	8	1814	-163.612453
0.100000	16	1500	-161.891517
0.100000	24	1430	-163.516832

Additional hidden nodes improve performance until there are 7 nodes in which, beyond which performance is flat. All models outperform the null model.

Reconstruction loss is minimized with η or .1 and N1 of 11. Hit rate and cross entropy loss actually improve for the appended autoencoder network as compared to the regular multilayer network, suggesting input compression silenced unimportant features. This effect is likely due to the large number of categorical features in the Car dataset.

4.4 Abalone

For the linear models, MSE is minimized with η .001, while only η of .1 does not outperform the null model.

Surprisingly, no combination of N1 and N2 outperforms the null model in the basic multilayer networks.

Table 8: House Votes 84 Autoencoder Appended to Network

η	N2	Samples	Loss	Hit Rate	Null Model	Autoencoder η	Autoencoder N
0.100000	8	4083	-161.571014	0.944424	0.613186	0.100000	8
0.100000	16	4432	-161.094018	0.941002	0.613186	0.100000	8
0.100000	32	3734	-157.955781	0.922072	0.613186	0.100000	8

Table 9: Car Linear Model

η	Samples	Loss	Hit Rate	Null Model
0.000100	27680	-349.385418	0.699422	0.699422
0.001000	15362	-431.468631	0.750000	0.699422
0.010000	10933	-509.416829	0.834393	0.699422
0.100000	6643	-558.869594	0.871821	0.699422

Autoencoder performance peaks for η of .01 and varies little with N1. When appended to a regression network, the model outperforms the null model for η of .01, minimizing loss for N2 of 9.

4.5 Forest Fires

None of the linear models for forest fires outperforms the null model, though mean squared error does bottom at η of .001.

Once again, no multilayer model outperforms the null, as each combination of N1 and N2 results in a mean squared error on the testing set higher than that for the null model.

Reconstruction loss for the autoencoders is minimized with η of .001. It does not vary significantly with N1. However, the appended autoencoder models also do not outperform the null.

4.6 Machine

As per Table 21, the linear models perform best with an η of .001, and they significantly outperform the null model in terms of mean squared error.

Interestingly, Table 22 shows that loss is minimized by higher numbers of nodes in the hidden layers, except when the first layer has 24 nodes. Loss is minimized when N1 is 12 and N2 is 12 or 24. All combinations outperform the null.

Loss is minimized for η of .001 and N1 of 3 or 4. I.e. the autoencoder needs more than 1 compression node. When appended to a regression network, the models outperform the null for N2 of 12 and 24.

5 Discussion

The time required to perform this tuning varied by dataset and network parameters and strongly correlated with both the number of samples and the number of nodes. The former is not a surprise given the training approach taken that tested model performance, and

Table 10: Car Multilayer Model

η	N1	N2	Samples	Loss	Hit Rate	Null Model
0.100000	3	3	3736	-407.013487	0.725867	0.699422
0.100000	3	7	6920	-552.269136	0.857659	0.699422
0.100000	3	15	6366	-548.104069	0.856503	0.699422
0.100000	3	30	7058	-548.675790	0.853757	0.699422
0.100000	3	60	6920	-554.482542	0.857225	0.699422
0.100000	7	3	7335	-513.852128	0.830202	0.699422
0.100000	7	7	6920	-552.402233	0.860694	0.699422
0.100000	7	15	6920	-556.186377	0.863728	0.699422
0.100000	7	30	5674	-545.985890	0.848555	0.699422
0.100000	7	60	6504	-559.105226	0.851301	0.699422
0.100000	15	3	5259	-441.744667	0.767919	0.699422
0.100000	15	7	7335	-558.784497	0.851012	0.699422
0.100000	15	15	7058	-555.054082	0.850000	0.699422
0.100000	15	30	7196	-561.324168	0.866763	0.699422
0.100000	15	60	6504	-558.098587	0.865029	0.699422
0.100000	30	3	5536	-445.677679	0.760405	0.699422
0.100000	30	7	7058	-536.641482	0.834249	0.699422
0.100000	30	15	7196	-559.235258	0.857948	0.699422
0.100000	30	30	7612	-562.896419	0.863439	0.699422

Table 11: Car Autoencoder

η	N1	Samples	Loss
0.010000	3	11072	-453.389336
0.010000	7	6781	-413.795128
0.010000	11	3460	-382.042615
0.100000	3	7196	-558.629705
0.100000	7	8719	-567.675872
0.100000	11	8442	-570.238747

therefore could only stop training, after every 2 updates for each sample in the training set. The latter's effect resulted from the recursive approach taken to backpropagation in the networks. For backpropagation in the multilayer networks, when a given node went to update its input weights given an error amount, it first recursively updated the weights for each of its parents, which, if they were in the second hidden layer, would recursively update their own parents' input weights. Since each node was a child of every node in the previous layer, the total number of updates made for a single backpropagation in a multilayer network is $o * N1 * N2$, where o is the number of output nodes. While O is fixed, N1 and N2 were varied for tuning, resulting in far longer testing times as they increased.

Table 12: Car Autoencoder Appended to Network

η	N2	Samples	Loss	Hit Rate	Null Model	Autoencoder η	Autoencoder N
0.100000	15	14532	-572.956642	0.873844	0.699422	0.100000	11
0.100000	30	13563	-565.795821	0.872110	0.699422	0.100000	11

Table 13: Abalone Linear Model

η	Samples	Loss	Null Model
0.000100	50464	0.546147	0.999402
0.001000	18715	0.506993	0.999402
0.010000	8020	0.538030	0.999402
0.100000	8689	1.465032	0.999402

If we were to perform this project again, we would change this update approach to aggregate updates from each layer, changing the backpropagation performance to $O(o + N1 + N2)$. Also, we would tune the frequency of performance checks during training.

6 Conclusion

Our neural network models significantly outperform the null models for 5 of the 6 datasets, except the multilayer models, which only outperform for 4 of 6. Accuracy on each dataset is optimized for a unique set of parameters, i.e. there is no single algorithm optimized to classify all of them. This finding is consistent with the no free lunch theorem. For all 6 datasets, autoencoding does not degrade performance and in one case even improved performance, allowing for a reduction in dimensionality and thereby improved performance. The exception to the outperformance for all 3 constructions is the forest fires dataset, for which no parameter set significantly outperforms the null model. One possible explanation for this observation is that the features provided in the dataset do not have predictive power for the burned area of the forest. Alternatively, neural networks may be an inappropriate model for this dataset, and another model may perform better with the same features.

Table 14: Abalone Multilayer Model

η	N1	N2	Samples	Loss	Null Model
0.001000	4	2	6684	0.999704	0.999402
0.001000	4	4	6684	0.999820	0.999402
0.001000	4	9	6684	1.000005	0.999402
0.001000	4	18	6684	1.001942	0.999402
0.001000	9	2	6684	1.000066	0.999402
0.001000	9	4	6684	0.999751	0.999402
0.001000	9	9	6684	1.000640	0.999402
0.001000	9	18	6684	1.004283	0.999402
0.001000	18	2	6684	0.999727	0.999402
0.001000	18	4	6684	0.999932	0.999402
0.001000	18	9	6684	1.000034	0.999402
0.001000	18	18	7018	1.000781	0.999402

Table 15: Abalone Autoencoder

η	N1	Samples	Loss
0.001000	2	6684	0.998131
0.001000	4	6684	0.995008
0.001000	6	26067	0.845850
0.010000	2	18715	0.487762
0.010000	4	18381	0.484278
0.010000	6	15373	0.493161

Table 16: Abalone Autoencoder Appended to Network

η	N2	Samples	Loss	Null Model	Autoencoder η	Autoencoder N
0.001000	2	40104	0.999563	0.999402	0.010000	4.000000
0.001000	9	37430	0.999814	0.999402	0.010000	4.000000
0.010000	2	45451	0.897980	0.999402	0.010000	4.000000
0.010000	9	52135	0.629826	0.999402	0.010000	4.000000

Table 17: Forest Fires Linear Model

η	Samples	Loss	Null Model
0.000100	6651	2.021823	1.839538
0.001000	1983	1.916511	1.839538
0.010000	908	2.051623	1.839538
0.100000	826	12297928681988.232422	1.839538

Table 18: Forest Fires Multilayer Model

η	N1	N2	Samples	Loss	Null Model
0.001000	6	6	1734	1.854377	1.841442
0.001000	6	13	1239	1.853309	1.841442
0.001000	6	27	1032	1.856690	1.841442
0.001000	6	54	949	1.859286	1.841442
0.001000	13	6	1652	1.853576	1.841442
0.001000	13	13	1363	1.854550	1.841442
0.001000	13	27	867	1.853185	1.841442
0.001000	13	54	991	1.874305	1.841442
0.001000	27	6	1693	1.853644	1.841442
0.001000	27	13	1239	1.855810	1.841442
0.001000	27	27	1032	1.856046	1.841442

Table 19: Forest Fires Autoencoder

η	N1	Samples	Loss
0.000100	6	5659	1.897402
0.000100	13	4296	1.868867
0.000100	20	3593	1.858362
0.001000	6	1610	1.847878
0.001000	13	1280	1.848311
0.001000	20	1239	1.849914

Table 20: Forest Fires Autoencoder Appended to Network

η	N2	Samples	Loss	Null Model	Autoencoder η	Autoencoder N
0.000100	6	5905	1.913815	1.837961	0.001000	6
0.000100	13	4749	1.885027	1.837961	0.001000	6
0.001000	6	2436	1.872259	1.837961	0.001000	6
0.001000	13	2023	1.871687	1.837961	0.001000	6

Table 21: Machine Linear Model

η	Samples	Loss	Null Model
0.000100	6216	11688.361449	28770.497548
0.001000	1192	9353.666232	28770.497548
0.010000	420	10253.709209	28770.497548
0.100000	487	540421153.171969	28770.497548

Table 22: Machine Multilayer Model

η	N1	N2	Samples	Loss	Null Model
0.001000	3	3	1075	20073.709224	28831.142560
0.001000	3	6	823	20579.885495	28831.142560
0.001000	3	12	739	18500.539153	28831.142560
0.001000	3	24	520	17444.899948	28831.142560
0.001000	6	1	1814	21859.402843	28831.142560
0.001000	6	3	1125	19710.516475	28831.142560
0.001000	6	6	907	18827.776709	28831.142560
0.001000	6	12	588	17776.420918	28831.142560
0.001000	6	24	470	19443.767192	28831.142560
0.001000	12	1	1797	21969.551819	28831.142560
0.001000	12	3	1327	19602.980657	28831.142560
0.001000	12	6	873	20015.870593	28831.142560
0.001000	12	12	789	16364.422986	28831.142560
0.001000	12	24	504	16353.354935	28831.142560
0.001000	24	1	1898	22841.537103	28831.142560
0.001000	24	3	1260	18955.315992	28831.142560
0.001000	24	6	840	19570.512585	28831.142560
0.001000	24	12	638	18371.438372	28831.142560
0.001000	24	24	470	18906.880944	28831.142560

Table 23: Machine Autoencoder

η	N1	Samples	Loss
0.001000	1	2217	21500.194794
0.001000	3	1680	17807.853276
0.001000	4	1293	17659.238136
0.010000	1	588	22078.680052
0.010000	3	453	19772.850616
0.010000	4	420	19359.745351

Table 24: Machine Autoencoder Appended to Network

η	N2	Samples	Loss	Null Model	Autoencoder η	Autoencoder N
0.001000	12	1008	18886.132691	28694.474872	0.001000	4.000000
0.001000	24	806	18941.069244	28694.474872	0.001000	4.000000